

A Rooms-and-Passages Domain with Discrete Steklov Spectrum but No Weyl Law

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Status. Candidate counterexample, likely valid, pending expert review. The proof is complete and has no numerical or unproved dependency.

1 Source question

Karpukhin, Lagacé, and Polterovich ask in Remark 4.3 of *Weyl’s law for the Steklov problem on surfaces with rough boundary* [1] whether there exist domains for which the Steklov spectrum is discrete but Weyl’s law does not hold. The remark appears on page 13 of the arXiv PDF. In exact transcription, the question is:

“It would be interesting to understand if there exist domains for which the Steklov spectrum is discrete but the Weyl’s law (1.2) does not hold.”

Proposition 4.2. *Let $\Omega \subset \mathbb{R}^2$ be a domain with piecewise smooth boundary, possibly with interior and exterior cusps. If all exterior cusps are slow then Weyl’s law (1.2) holds for the counting function of the Steklov eigenvalues on Ω . Moreover, if Ω has at least one fast cusp, then the Steklov spectrum of Ω is not discrete.*

Remark 4.3. It would be interesting to understand if there exist domains for which the Steklov spectrum is discrete but the Weyl’s law (1.2) does not hold. To construct such an example one needs to find a domain Ω with the boundary conformal factor in $L^1 \setminus L \log L$, and yet for which the resolvent of the Dirichlet-to-Neumann map is still compact. We note that in terms of the weighted problem it seems like this would require going beyond the Orlicz scale: indeed, for every $0 \leq a < 1$ one can find $\beta \in L(\log L)^a(\partial\Omega)$ so that the embedding $W^{1,2}(\partial\Omega) \rightarrow L^2(\partial\Omega, \beta d\ell)$ is not compact, following the proof found in [GKL21 Example 3.19].

For a planar domain with finite boundary length, the asserted Weyl law is

$$N_\Omega(\sigma) = \frac{|\partial\Omega|}{\pi} \sigma + o(\sigma), \quad \sigma \rightarrow \infty, \tag{1}$$

where $N_\Omega(\sigma) = \#\{k : \sigma_k(\Omega) < \sigma\}$.

We use the natural variational setup of the source paper. The boundary Sobolev space $H_\partial^1(\Omega)$ is the completion of the finite-norm piecewise-smooth functions under

$$\|u\|_{H_\partial^1(\Omega)}^2 = \int_\Omega |\nabla u|^2 dx + \int_{\partial\Omega} |u|^2 ds. \tag{2}$$

Its boundary map into $L^2(\partial\Omega)$ is denoted by Tr . When this map is compact, the associated Steklov form has compact resolvent, and its eigenvalues have the min–max characterization

$$\sigma_k(\Omega) = \inf_{\dim E=k+1} \sup_{0 \neq u \in E} \frac{\int_{\Omega} |\nabla u|^2 dx}{\int_{\partial\Omega} |\text{Tr } u|^2 ds}. \quad (3)$$

2 Proof intuition

Attach infinitely many tiny rooms to a square through much thinner passages. A function supported in one room pays only the passage conductance w_j/r_j in Dirichlet energy, while it receives boundary mass of order r_j from the room walls. Thus its Steklov quotient is roughly w_j/r_j^2 . We choose this to be merely $\log j$, producing far too many low Steklov eigenvalues for a linear counting law.

The passages cannot be made arbitrarily thin without losing compactness of the trace. Two effects govern the exact threshold. A neck of width w_j leading to a room of size r_j amplifies inlet boundary mass by r_j/w_j and energy by r_j^2/w_j . Meanwhile an interval of length w_j on the square carries at most $Cw_j \log(e/w_j)$ of the L^2 mass of a bounded $H^{1/2}$ trace. With

$$r_j = e^{-j^2}, \quad w_j = r_j^2 \log(j+1),$$

the inlet tail is summable and the energy amplification tends to zero. This is exactly enough for compactness, while the room test functions retain their logarithmic quotients.

3 The counterexample

Let $Q = (0, 1)^2$. For $j \geq 3$, set

$$x_j = 2^{-j}, \quad r_j = e^{-j^2}, \quad w_j = r_j^2 \log(j+1). \quad (4)$$

Above the interval

$$B_j = (x_j - w_j/2, x_j + w_j/2) \times \{1\}$$

attach the passage and room

$$\begin{aligned} P_j &= (x_j - w_j/2, x_j + w_j/2) \times (1, 1 + r_j), \\ R_j &= (x_j - r_j/2, x_j + r_j/2) \times (1 + r_j, 1 + 2r_j). \end{aligned}$$

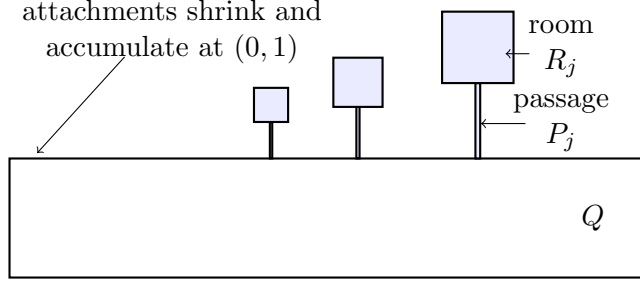
Precisely, define

$$\Omega = \text{int} \left(\overline{Q} \cup \bigcup_{j \geq 3} (\overline{P}_j \cup \overline{R}_j) \right). \quad (5)$$

The pieces are mutually disjoint away from their intended seams: indeed $r_j = o(2^{-j})$. They accumulate only at $(0, 1)$, and their heights tend to zero. Following the top boundary of Q and detouring around each rectangular attachment therefore gives a simple closed curve. Moreover,

$$|\partial\Omega| \leq 4 + C \sum_{j \geq 3} r_j < \infty.$$

Thus Ω is a bounded simply connected planar domain with rectifiable Jordan boundary.



We next prove compactness of the trace. Write $A_j = P_j \cup R_j$ and let $\Gamma_j = \partial\Omega \cap \partial A_j$ be the external boundary of the j th attachment.

Lemma 1 (Critical interval estimate). *There is a constant C such that, for every interval I in one edge of ∂Q with length $0 < \delta \leq 1/2$ and every $v \in H^{1/2}(\partial Q)$,*

$$\int_I |v|^2 ds \leq C\delta \log(e/\delta) \|v\|_{H^{1/2}(\partial Q)}^2. \quad (6)$$

Proof. The one-dimensional critical Sobolev estimate gives, for every $p \geq 2$,

$$\|v\|_{L^p(\partial Q)} \leq C\sqrt{p} \|v\|_{H^{1/2}(\partial Q)}. \quad (7)$$

For completeness, parameterize the whole polygonal curve ∂Q by arclength as a circle. Hausdorff–Young and Cauchy–Schwarz then give (7): the relevant weight sum is $\sum_n (1 + |n|)^{-p/(p-2)} = O(p)$, and its resulting exponent is at most $1/2$. Hölder’s inequality now yields

$$\int_I |v|^2 ds \leq \delta^{1-2/p} \|v\|_{L^p}^2 \leq Cp\delta^{1-2/p} \|v\|_{H^{1/2}}^2.$$

Take $p = 2 \log(e/\delta)$; then $\delta^{-2/p}$ is bounded by an absolute constant. This proves (6). $\square$

Lemma 2 (Uniform neck trace estimate). *Let a square room of side r be joined to an inlet interval B by a rectangular passage of length r and width w , where $0 < w \leq r/2$. If A is the room–passage union and Γ is its external boundary, then every $u \in H^1(A)$ satisfies*

$$\int_{\Gamma} |u|^2 ds \leq C \left(\frac{r}{w} \int_B |u|^2 ds + \frac{r^2}{w} \int_A |\nabla u|^2 dx \right), \quad (8)$$

with an absolute constant C .

Proof. Scale by r and put $\eta = w/r$. The passage becomes $C_\eta = (0, \eta) \times (0, 1)$, with bottom inlet B and top opening T . The one-dimensional fundamental theorem of calculus on each vertical line gives

$$\int_{C_\eta} |u|^2 \leq 2 \int_B |u|^2 + 2 \int_{C_\eta} |\partial_y u|^2. \quad (9)$$

The one-dimensional trace inequality on each horizontal cross-section, integrated in y , bounds the two vertical sides by

$$\frac{C}{\eta} \int_{C_\eta} |u|^2 + C\eta \int_{C_\eta} |\partial_x u|^2.$$

The same vertical fundamental-theorem estimate bounds $\int_T |u|^2$ by $2 \int_B |u|^2 + 2 \int_{C_\eta} |\partial_y u|^2$. Consequently the passage boundary and its top opening are bounded by

$$\frac{C}{\eta} \left(\int_B |u|^2 + \int_{C_\eta} |\nabla u|^2 \right). \quad (10)$$

For the unit square room S , let m be the mean of u on S . Standard Poincaré and trace inequalities give

$$\|u - m\|_{L^2(\partial S)}^2 \leq C \int_S |\nabla u|^2.$$

Comparing m with the mean of u on the opening T , whose length is η , gives

$$|m|^2 \leq \frac{C}{\eta} \int_T |u|^2 + \frac{C}{\eta} \int_S |\nabla u|^2.$$

It follows that the room boundary is bounded by the right side of (10), with the room energy added. Hence, at unit scale,

$$\int_\Gamma |u|^2 \leq \frac{C}{\eta} \left(\int_B |u|^2 + \int_A |\nabla u|^2 \right).$$

Boundary measure scales by r , whereas two-dimensional Dirichlet energy is scale invariant. Multiplying the last inequality by r and using $\eta = w/r$ gives (8). $\square$

Lemma 3 (Compact trace). *The boundary embedding*

$$\text{Tr} : H_\partial^1(\Omega) \longrightarrow L^2(\partial\Omega)$$

is compact. In particular, the natural Steklov spectrum of Ω is discrete.

Proof. We first work on the dense piecewise-smooth core. The bottom edge of Q is part of $\partial\Omega$. Poincaré's inequality anchored on that edge and the ordinary trace theorem on Q imply

$$\|u|_Q\|_{H^1(Q)}^2 + \|u|_{\partial Q}\|_{H^{1/2}(\partial Q)}^2 \leq C \|u\|_{H_\partial^1(\Omega)}^2. \quad (11)$$

Apply Lemmas 1 and 2 to every attachment with $j \geq J$. Since the B_j are disjoint,

$$\begin{aligned} \sum_{j \geq J} \int_{\Gamma_j} |u|^2 \, ds &\leq C \sum_{j \geq J} \frac{r_j}{w_j} \int_{B_j} |u|^2 \, ds + C \sum_{j \geq J} \frac{r_j^2}{w_j} \int_{A_j} |\nabla u|^2 \, dx \\ &\leq C \left(\sum_{j \geq J} r_j \log(e/w_j) + \sup_{j \geq J} \frac{r_j^2}{w_j} \right) \|u\|_{H_\partial^1(\Omega)}^2. \end{aligned} \quad (12)$$

For the parameters (4),

$$\sum_{j \geq J} r_j \log(e/w_j) \longrightarrow 0, \quad \sup_{j \geq J} \frac{r_j^2}{w_j} = \frac{1}{\log(J+1)} \longrightarrow 0.$$

Thus the L^2 trace outside any finite collection of attachments has uniformly vanishing operator norm.

It remains to combine this with local compactness. A sequence bounded in $H_\partial^1(\Omega)$ is bounded in $H^1(Q)$ by (11). It is also bounded in $H^1(A_j)$ for each fixed j , by the same Poincaré argument

anchored on any fixed external side of A_j (whose L^2 trace is part of (2)). The ordinary compact trace theorem on these finitely many polygonal pieces therefore gives a subsequence whose traces converge on the base boundary and on $\Gamma_3, \dots, \Gamma_{J-1}$. Estimate (12) makes the remaining trace uniformly small. Hence this subsequence is Cauchy in $L^2(\partial\Omega)$.

All estimates are stable under the norm (2), so they extend from the dense core to $H_{\partial}^1(\Omega)$. This proves compactness. The standard closed-form construction, equivalently the variational setup (3) used in [1], then gives compact resolvent and a discrete Steklov spectrum. $\square$

Theorem 1. *The domain Ω in (5) has discrete Steklov spectrum, but its counting function satisfies*

$$\limsup_{\sigma \rightarrow \infty} \frac{N_{\Omega}(\sigma)}{\sigma} = \infty. \quad (13)$$

In particular, the Weyl law (1) fails. This answers Remark 4.3 of [1] affirmatively.

Proof. Discreteness follows from Lemma 3. For $j \geq 3$, define u_j to be zero on Q and on every other attachment, to increase linearly from 0 to 1 along the vertical coordinate of P_j , and to equal 1 on R_j . These continuous piecewise-affine functions belong to the natural boundary Sobolev space (or may be approximated there by smooth functions). Their Dirichlet energies are

$$\int_{\Omega} |\nabla u_j|^2 dx = w_j r_j \frac{1}{r_j^2} = \frac{w_j}{r_j} = r_j \log(j+1). \quad (14)$$

The external boundary of the room has length $4r_j - w_j$ and carries value 1. The two passage sides contribute an additional $2 \int_0^{r_j} (t/r_j)^2 dt = 2r_j/3$. In particular,

$$\int_{\partial\Omega} |u_j|^2 ds \geq 3r_j, \quad \frac{\int_{\Omega} |\nabla u_j|^2}{\int_{\partial\Omega} |u_j|^2} \leq \log(j+1). \quad (15)$$

The functions have disjoint supports, so both the energy and boundary quadratic forms are diagonal on their span. The N -dimensional space $E_N = \text{span}\{u_3, \dots, u_{N+2}\}$ therefore satisfies

$$\sup_{0 \neq u \in E_N} \frac{\int_{\Omega} |\nabla u|^2}{\int_{\partial\Omega} |u|^2} \leq \log(N+3).$$

The min-max formula (3) gives

$$\sigma_{N-1}(\Omega) \leq \log(N+3). \quad (16)$$

Consequently, for $\tau_N = 2 \log(N+3)$ and all sufficiently large N ,

$$N_{\Omega}(\tau_N) \geq N, \quad \frac{N_{\Omega}(\tau_N)}{\tau_N} \geq \frac{N}{2 \log(N+3)} \rightarrow \infty.$$

This proves (13). Since $0 < |\partial\Omega| < \infty$, it contradicts the finite limit $|\partial\Omega|/\pi$ required by (1). $\square$

4 Verification notes and limitations

The construction satisfies every hypothesis in the source question: Ω is a genuine bounded planar domain, its boundary is a finite-length Jordan curve, and its natural Steklov resolvent is compact by Lemma 3. The exact Weyl conclusion fails by the quantitative lower bound (13); the failure is stronger than a wrong leading constant.

No computational assertion or unproved lemma enters the proof. The two review-sensitive points are the uniform scaling in Lemma 2 and the extension of the tail estimate to the completed boundary Sobolev space. Both are written out explicitly. The packet does not claim Weyl failure for ordinary $H^1(\Omega)$ under an alternative trace convention; it uses precisely the natural rough-domain variational space selected in the source paper.

5 Novelty check

The run’s registry, solution, attempt, and proof-gap indexes were searched for the arXiv id, source title, Steklov discreteness without Weyl asymptotics, and rooms-and-passages terminology. No duplicate was found. Bounded web searches on 9 August 2026 used the exact sentence from Remark 4.3 and variants of **Steklov spectrum discrete Weyl law counterexample** and **rooms and passages Steklov**. They found the source paper and classical rooms-and-passages examples for other operators, but no later resolution of Remark 4.3 or matching Steklov example. This establishes only bounded-search novelty, not an exhaustive literature guarantee.

6 Human-review recommendation

Review Lemmas 2 and 3 first. If their trace-space implementation is accepted, the disjoint test functions and min–max contradiction are elementary, and the packet should be accepted as a full affirmative construction answering Remark 4.3.

References

- [1] M. Karpukhin, J. Lagacé, and I. Polterovich, *Weyl’s law for the Steklov problem on surfaces with rough boundary*, arXiv:2204.05294 (2022), Remark 4.3.